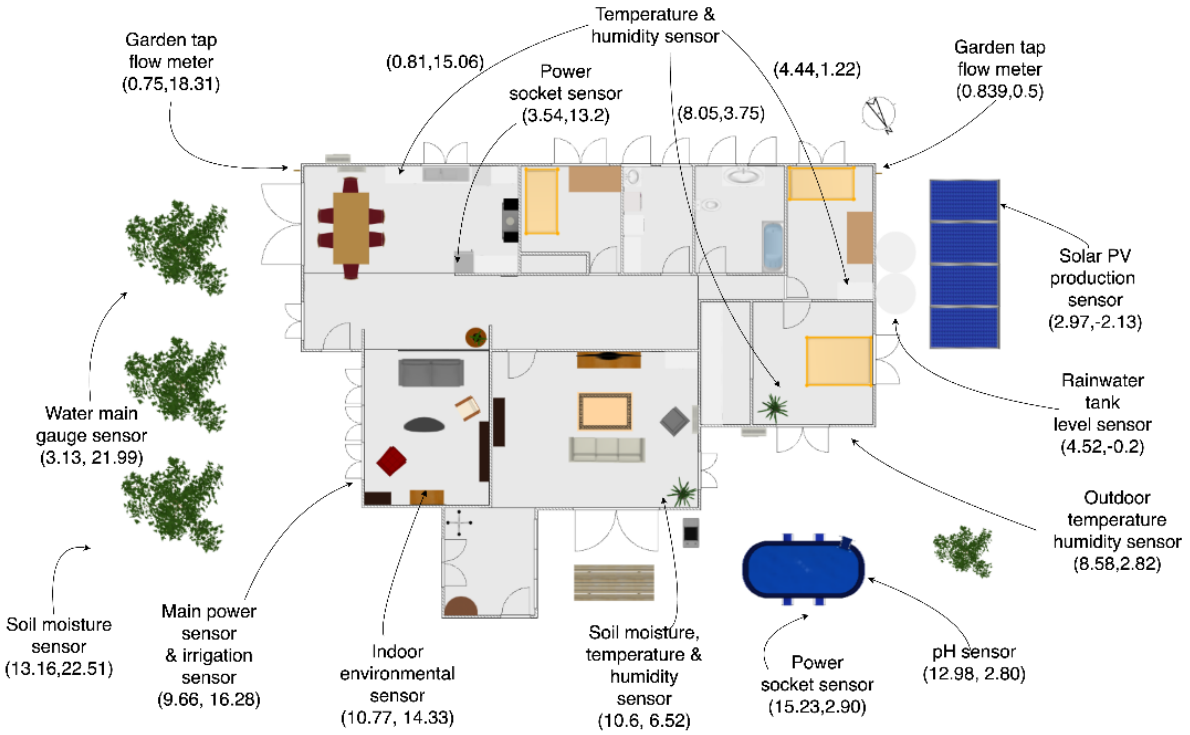


Figure 3.2 Using a floor plan to select the optimal placement of sensors based on what parameters need to be measured, as well as the location of power outlets and taps. The local coordinates of each sensor relative to the origin point (close to the compass symbol at the top right of the image) in meters are also shown.

3.2.3 Determine required performance

Match sensor capabilities to what your digital twin actually needs to deliver value and resist the urge to specify the highest performance possible. For accuracy, ask yourself what decisions the data will drive. My home's temperature sensors only need ±0.5°C accuracy because my HVAC system can't control more precisely than that anyway. Paying extra for ±0.1°C accuracy would be wasting money on precision I can't use. Similarly, sampling frequency should match your system's dynamics and control capabilities. While my pool pump could report power consumption every second, I only sample hourly because that's sufficient to detect when it's running and calculate daily energy usage. More frequent sampling would just drain batteries faster without enabling better decisions. *Latency*, the delay between measurement and availability in your digital twin, matters most for real-time control. My irrigation system can tolerate long delays because soil moisture changes slowly, but a leak detection system needs sub-minute response to prevent damage. Understanding these requirements early prevents